INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

HYDROGEN LEAK DETECTOR

The General Services Administration has authorized the use of this commercial item description (CID), for all federal agencies.

1. SCOPE. This CID describes a portable hydrogen leak detector, which is used to determine the location of leaks in fuel tanks as well as other aircraft systems. By filling a fuel tank or container with hydrogen (the hydrogen acts as a tracer and is in a maximum concentration of 5% with 95% nitrogen), the hydrogen leak detector will enable aircraft maintainers to locate the origination point(s) of the leak(s) as the hydrogen escapes into the atmosphere.

2. SALIENT CHARACTERISTICS.

2.1 Hydrogen leak detector description. The hydrogen leak detector shall be supplied with a hydrogen leak detector unit, a probe with cable and associated connectors, battery charger, calibrated reference leak, and electrical cable to use 110VAC 50/60 Hz.

2.2 Design and construction. The hydrogen leak detector shall be designed and constructed so that no parts will work loose in service. They shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. They shall be weatherproof and designed to prevent the intrusion of water and sand into critical operating components. The detector shall display leak rates in units of parts per million (ppm).

2.2.1 Electromagnetic interference (EMI). The hydrogen leak detector shall be in accordance with the following radiated emission and susceptibility requirements of MIL-STD-461: RE102 and RS103.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data that may improve this document should be sent to WR-ALC/403 SCMS, 460 Richard Ray Blvd, Suite 200, Robins AFB, GA 31098-1813. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

AMSC N/A FSC 6635

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2.2.2 Electrostatic discharge. The design of the hydrogen leak detector shall preclude equipment damage due to electrostatic discharge (ESD), protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking.

2.2.3 Human engineering. The hydrogen leak detector shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens and Mission-Oriented Protective Posture (MOPP) Level 4 Chemical Warfare Gear.

2.3 Environmental conditions.

2.3.1 Operating temperature range. The hydrogen leak detector shall be capable of operating in ambient temperatures ranging from 0ºC to 50ºC.

2.3.2 Operating conditions. The hydrogen leak detector shall be authorized for operation in a Class I, Division 1 environment where ignitable concentrations of flammable gases, vapors, or liquids are present within the atmosphere under normal operational conditions in accordance with UL 913, UL 1203, NFPA 496, and UL 60079-11.

2.3.3 Certificate of conformance. The hydrogen leak detector shall be supplied with a certificate of conformance certifying compliance with UL 913 Class I, Division 1 requirements.

2.4 Performance.

2.4.1 Concentration. The minimum hydrogen detectable concentration required shall be 0.5 parts per million.

2.4.2 Calibration. The hydrogen leak detector shall have an internal calibration function that can be performed by the field users.

2.5 Standard probe. The detector shall be supplied with a handheld replaceable hydrogen detecting probe with cable and associated connectors.

2.6 Input voltage source. The leak detector shall operate at 110 or 220 VAC, 50 to 60 Hz, as well as 12 volt direct current (VDC) at 770 milliamps using a rechargeable lithium ion battery or equivalent.

2.7 Batteries. At least one commercially available lithium ion battery (or equivalent) shall be supplied as the power source for detector operation. The hydrogen leak detector shall have a run time of at least 5 hours before having to re-charge the battery.

2.7.1 Battery charger. The hydrogen leak detector shall be supplied with a battery charger to charge its battery to a fully charged state in less than 12 hours. A fully charged state is defined as 12.6 volts and 770 milliamps.

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2.8 Foreign power requirements. When foreign power sources are utilized due to the leak detectors being operated at foreign destinations, proper electrical plugs/adapters and transformers shall be provided to allow the leak detector unit and leak detector battery charger to operate properly.

2.9 Operating conditions.

2.9.1 Relative humidity. The hydrogen leak detector shall be capable of operating at a maximum relative humidity of 90%.

2.10 Detector dimensions. The dimensions of the hydrogen leak detector shall not exceed 5.25in x 9.5in x 6.75in.

2.11 Detector weight. The hydrogen leak detector shall not weigh more than eleven pounds including the battery.

2.12 Storage/carrying case. The hydrogen leak detector shall be supplied with a storage/carrying case capable of storing the detector and all related components including the operation manual.

2.13 Workmanship. The hydrogen leak detector, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

2.13.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

2.13.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.0 REGULTORY REQUIREMENTS

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4.0 PRODUCT CONFORMANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of the CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be appropriate. Proof of conformance may include, but shall not be limited to the performance of operational tests, lab tests, modeling and simulation and delivery of reports and data from these tests.

4.2 First production inspection. The first production hydrogen leak detector shall be subjected to the analyses, demonstrations, examinations, and tests described in 4.4.1 through 4.4.9. The contractor shall provide or arrange for all test equipment and facilities.

4.3 Inspection requirements.

4.3.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity

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4.3.3 Test rejection criteria. Throughout all tests specified herein, the hydrogen leak detector shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Failure to conform to UL 913 Class I, Division 1 requirements.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Misalignment of components.

f. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

g. Evidence of corrosion or deterioration.

4.4 Test methods.

4.4.1 Examination of product. Each hydrogen leak detector shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 4.4. A contractor-generated, Government-approved checklist shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each hydrogen leak detector function shall be verified. Certifications and analyses shall be provided in accordance with Table I.

4.4.2 Electromagnetic interference test. A first production hydrogen leak detector shall be tested in accordance with MIL STD 461: RE 102 and RS 103 to demonstrate compliance with 2.2.1.

4.4.3 Maintainability demonstration. All recommended preventive maintenance tasks shall be performed and the task times shall be recorded. The recommended frequencies of the preventive maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of preventive maintenance time per measure of use, such as calendar time or hours of operation. It shall be demonstrated that the forces required do not exceed those allowed in MIL STD 1472. All preventive maintenance tasks recommended to be performed daily and at the routine PMI shall also be performed by personnel wearing arctic mittens and MOPP Level 4 Chemical Warfare Gear.

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TABLE I. Certifications and analyses.

Paragraph	Required Certifications and Analyses
2.2.2 Electrostatic discharge.	Contractor certification that the hydrogen leak detector prevents electrostatic discharge (ESD) requirements as defined in 2.2.2.
2.3.3Certificate of conformance.	Contractor certification that the hydrogen leak detector certified for use in a Class I Division 1 environment as defined in 2.3.2.
2.6 Input voltage source.	Contractor analysis of the input voltage requirement: An engineering analysis shall be performed to demonstrate compliance with the input voltage requirements of 2.6.
2.7 Batteries.	Contractor analysis of the batteries requirement: An engineering analysis shall be performed to demonstrate compliance with the batteries requirements of 2.7.
2.7.1 Battery charger.	Contractor analysis of the battery charger requirement: An engineering analysis shall be performed to demonstrate compliance with the battery charger requirements of 2.7.1.

4.4.4 Operating temperature range test. A first production hydrogen leak detector shall be tested to demonstrate compliance with 2.3.1.

4.4.5 Concentration test. A first production hydrogen leak detector shall be tested to demonstrate compliance with 2.4.1

4.4.6 Calibration test. A first production hydrogen leak detector shall be tested to demonstrate compliance with 2.4.2.

4.4.7 Relative humidity test. A first production hydrogen leak detector shall be tested to demonstrate compliance with 2.9.1.

4.4.8 Detector dimensions test. A first production hydrogen leak detector shall be measured to demonstrate compliance with the dimensional requirements of 2.10.

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4.4.9 Detector weight test. A first production hydrogen leak detector shall be weighed and value recorded to demonstrate compliance with 2.11.

5. PACKAGING.

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Intended use. The purpose of the hydrogen leak detector is to locate leaks that are often found within tanks or containers. This will allow the aircraft maintainers to take corrective action and patch or fill the leaking areas.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this commercial item description.

b. If additional electrical plugs/adapters, and transformers are required to interface with foreign power sources (see 2.8).

6.3 Source of documents.

6.3.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at https://assist.dla.mil or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.3.2 FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/ . Electronic copies of the DFARS may be obtained from http://www.acq.osd.mil/dpap/dars/dfars/ index.htm.

6.3.3 National Fire Protection Association (NFPA)/National Electrical Code (NEC) standards may be obtained at http://www.nfpa.org or available from 1 Batterymarch Park, Quincy, Massachusetts USA 02169-7471

6.3.4 Underwriters Laboratories (UL) copies may be obtained online at http://ulstandardsinfonet.ul.com/ or from: COMM 2000, 1414 Brook Drive, Downers Grove, IL 60515-5000.

6.4 Key words.

Locator
Portable
Tracer

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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

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